

LACS — Linear Analysis of Chemical Shifts

Detecting & Correcting NMR Referencing Errors

Tobias Senoner

22 April 2026

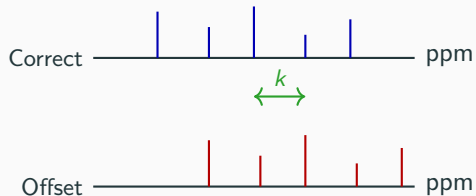
TriZOD Project Meeting

The Problem: Referencing Errors

Every NMR chemical shift is measured **relative to a reference standard** (e.g. DSS for ^{13}C).

If the reference is wrong by a constant k :

$$\delta_{\text{obs}} = \delta_{\text{true}} + k \quad \forall \text{ atoms of that nucleus type}$$



Prevalence in the BMRB:

- $\sim 20\%$ of ^{13}C entries mis-referenced
- $\sim 35\%$ of ^{15}N entries mis-referenced

Undetected offsets bias every downstream analysis.

Wang et al. (2005) *J Biomol NMR* 32:13–22; Wang & Markley (2009) *J Biomol NMR* 44:95–99

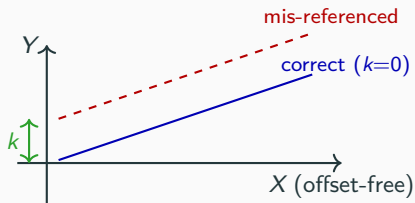
The Core Insight: Offset Cancellation

The **difference** of two ^{13}C atoms is immune to the offset:

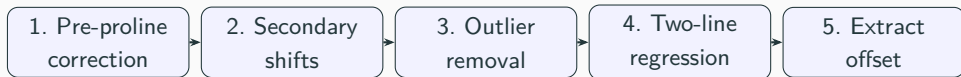
$$(CA_{\text{obs}} - CB_{\text{obs}}) = (CA_{\text{true}} + k) - (CB_{\text{true}} + k) = CA_{\text{true}} - CB_{\text{true}}$$

LACS strategy:

1. **X-axis:** secondary shift of the CA–CB *difference* (offset-free)
2. **Y-axis:** secondary shift of CA (retains offset k)
3. Non-zero **Y-intercept** $= k$



LACS Algorithm Overview (^{13}C / $^1\text{H}_\alpha$)



Applies to:

- CA, CB, C' (CO), H_α (HA)
- Excludes: Cys, Gly, Pro
- Requires ≥ 20 valid residues

Reference tables:

- Random coil shifts (Wishart 1995)
- Pre-proline corrections (20 AA)
- Thresholds per atom type

Step 1: Pre-Proline Correction

Proline's cyclic side chain systematically perturbs shifts of the **preceding residue**.

Before computing secondary shifts, add amino-acid-specific corrections to residue i when residue $i+1$ is Pro:

$$\delta_i^{\text{corr}} = \delta_i^{\text{obs}} + \Delta_{\text{pre-Pro}}(\text{AA}_i, \text{atom})$$

AA	ΔCA	ΔCB	ΔHA	ΔCO	ΔH	ΔN
Ala	+2.0	+1.0	-0.30	+1.9	+0.05	-1.2
Val	+2.4	+0.3	-0.32	+1.4	+0.01	-1.3
Leu	+2.0	+0.7	-0.34	+1.9	+0.02	-0.8
...	...	...	...	...	...	...

Purpose: Remove this known systematic effect so it doesn't contaminate the offset estimate.

Step 2: Secondary Shifts — X and Y Axes

For each valid residue i with amino acid type a :

X-axis (reference-independent):

$$X_i = \underbrace{(CA_i^{\text{obs}} - CB_i^{\text{obs}})}_{\text{observed diff.}} - \underbrace{(CA_a^{\text{rc}} - CB_a^{\text{rc}})}_{\text{random coil diff.}}$$

Any ^{13}C referencing error k cancels: $(x+k) - (y+k) = x - y$.

Y-axis (retains the offset):

$$Y_i = \text{target}_i^{\text{obs}} - \text{target}_a^{\text{rc}} \quad (\text{target} = \text{CA, CB, CO, or HA})$$

Worked Example: Val-15 in bmr10260

$$CA_{\text{obs}} = 58.57, \quad CB_{\text{obs}} = 34.15, \quad CA_{\text{rc}} = 62.2, \quad CB_{\text{rc}} = 32.9$$

$$X = (58.57 - 34.15) - (62.2 - 32.9) = -4.88$$

$$Y_{\text{CA}} = 58.57 - 62.2 = -3.63$$

The V-Shape: Secondary Structure Creates Two Branches

The CA—CB secondary shift correlates with secondary structure:

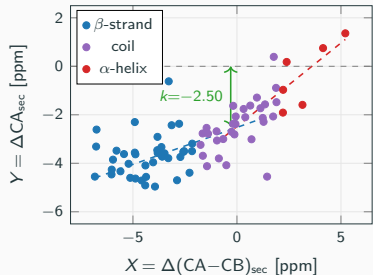
Helix $X \gg 0$
(high CA, low CB)

Coil $X \approx 0$

Strand $X \ll 0$
(low CA, high CB)

The Y-axis ($\Delta\delta$ of any single atom) has a **different slope** vs. X for helix vs. strand residues.

⇒ **Two-line fit** to capture both branches.



Step 3: Outlier Removal (Mahalanobis Distance)

Before regression, statistical outliers are removed:

1. Compute the **Mahalanobis distance** d_i of each (X_i, Y_i) from the centroid using the sample covariance matrix:

$$d_i = \sqrt{(\mathbf{x}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})}$$

2. Remove points where $d_i > 0.80 \times d_{\max}$
3. Cap at 2% of the dataset

Catches:

- Residues near paramagnetic centres or ligand-binding sites
- Assignment errors
- Unusual disulfide bonding effects

In our example (bmr10260): 82 $\rightarrow$ 77 points after outlier removal.

Step 4: Two-Line Robust Regression

Two **overlapping** regression lines, split by an atom-specific threshold:

Line	Points used	Captures	Threshold
Line 1 (strand side)	$X < +\text{thr}$	coil + β -strand	1 ppm (CA, CB)
Line 2 (helix side)	$X > -\text{thr}$	coil + α -helix	6 ppm (HA, C')

Each line is fitted with **robust regression** (IRLS, bisquare weights):

- Initialise with OLS
- Robust scale: $s = \text{MAD}(|r|)/0.6745$
- Bisquare weights: $w_i = (1 - u_i^2)^2$ if $|u_i| < 1$, else 0
- Re-fit WLS, iterate until convergence

Fallback: If either side has < 10 points, a single robust regression is used.

Step 5: Extract Offset

The referencing offset is the **mean of the two Y-intercepts**:

$$k = \frac{\beta_{0,\text{strand}} + \beta_{0,\text{helix}}}{2}$$

Why average two intercepts?

- Reduces sensitivity to the slope, which varies with protein composition
- Implicit quality check: large disagreement $\Rightarrow$ problematic data

Result for bmr10260

Strand-side intercept: $\beta_0 = -2.521$

Helix-side intercept: $\beta_0 = -2.483$

Offset: $k = \frac{-2.521 + (-2.483)}{2} = -2.50 \text{ ppm}$

$\Rightarrow$ All ^{13}C shifts in this entry are **2.50 ppm too low**. Subtract k (i.e. add 2.50) to correct.

Real Example: bmr10260

Protein: Fn3 domain of receptor-type tyrosine-protein phosphatase F (106 res.)

Atom	Offset (ppm)	Nucleus	Interpretation
CA	−2.50	^{13}C	Shared ^{13}C reference off by −2.5 ppm
CB	−2.50	^{13}C	
C'	see below	^{13}C	Same ^{13}C reference
HA	see below	^1H	^1H reference
HB	= HA	^1H	(propagated from HA)
H	see below	^1H	Amide ^1H
N	see below	^{15}N	^{15}N reference

Key property: CA and CB offsets are always (near-)identical because they share the same ^{13}C reference. This is both expected and a built-in sanity check.

Extension: ^{15}N and $^1\text{H}_\text{N}$ Offsets

Uses a **different linear relationship**: secondary shift of N_i correlates with CA—CB secondary shift of the **preceding residue** $i-1$.

$$X_i = (\text{CA} - \text{CB})_{i-1}^{\text{sec}}, \quad Y_i = \text{N}_i^{\text{sec}} - \text{Ncorr}(\text{AA}_{i-1})$$

Key differences from ^{13}C :

- **Inter-residue** correlation (backbone conformational coupling)
- **Preceding-residue correction** (Ncorr table, 20 AA types)
- **Two-phase outlier removal**: Mahalanobis + weight-based trimming
- **Constrained slope**: ≈ -0.4 (N) or ≈ -0.07 (HN); applied for small/unbalanced data
- **Systematic correction**: +0.465 ppm (N), +0.049 ppm (HN)

Additional filters: Pro (no amide), Gly/His as preceding (poor X predictors), $|X| \leq 10$

Sign Convention & Applying Corrections

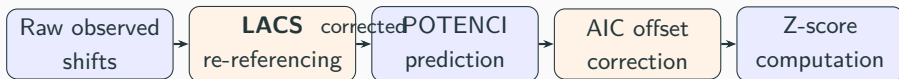
Convention:

offset $> 0 \Rightarrow$ observed too high $\Rightarrow$ subtract to correct: $\delta_i^{\text{corr}} = \delta_i^{\text{obs}} - k$

Atom type coverage:

Atom	Method	Notes
CA, CB	^{13}C regression	Same offset (shared reference)
C' (CO)	^{13}C regression	Wider threshold (6 ppm)
HA	^{13}C regression	Wider threshold (6 ppm)
HB	<i>propagated</i>	Set = HA offset (^1H nucleus)
N	^{15}N regression	Preceding-residue method
H (HN)	^{15}N regression	Preceding-residue method

LACS in the TriZOD Pipeline



Two complementary systems:

1. **LACS** (before POTENCI): Wishart 1995 random coil tables. Works on all residues.
2. **POTENCI-based offset** (after): POTENCI predictions + AIC model selection + 9-residue rolling window.

Status: LACS module complete & validated against 6,774 BMRB LACS reports. Integration into scoring pipeline in progress.

Limitations

- **Requires secondary shift spread.** IDPs with near-zero secondary shifts $\Rightarrow$ poorly determined regression. Less relevant for TriZOD (most BMRB entries are structured).
- **CA and CB offsets are coupled.** Both share the ^{13}C reference $\Rightarrow$ always near-identical offsets. Cannot detect atom-type-specific errors.
- **HB is inferred, not measured.** No direct LACS relationship for HB; offset propagated from HA.
- **Assumes constant offset.** Single constant per nucleus type. Position-dependent errors (rare) are not detected.

Summary

1. NMR referencing errors are **common** ($\sim 20\%$ ^{13}C , $\sim 35\%$ ^{15}N) and bias downstream analyses.
2. LACS detects offsets using the **reference-independent** CA–CB difference as X-axis.
3. Two-line robust regression handles the **V-shape** from helix/strand residues.
4. The **Y-intercept** directly gives the referencing error (sign: positive = too high).
5. Our Python reimplementation validates against BMRB LACS reports and serves as the first correction step in the TriZOD pipeline.